14 February 1975
SUPERSEDING
MIL-M-12942E
24 November 1971

MILITARY SPECIFICATION

# MOUNTS, 106MM RECOILLESS RIFLE: M79 AND M92 WITH TRIPOD: M27

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers two types of mounts for 106MM Recoilless Rifle.

1.2 Classification. Mounts shall be of the following types:

Type I - Mount, 106MM Recoilless Rifle: M79
Type II - Mount, 106MM Recoilless Rifle: M92 with Tripod: M27

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-13931 - Cannons, General Specification for
MIL-P-14232 - Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-G-21164 - Grease, Molybdenum Disulfide (For Low and High Temperatures)
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance, and Disposition of



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STANDARDS

Military

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage

PUBLICATIONS

U. S. Army Armament Command

EPL 8403200 - Engineering Parts List for Mount, 106MM Recoilless Rifle: M79
EPL 8407810 - Engineering Parts List for Tripod: M27
EPL 8407830 - Engineering Parts List for Mount, 106MM Recoilless Rifle: M92
LSQAP 8403200 - List of Supplementary Quality Assurance Provisions, Mount, 106MM Rifle: M79
LSQAP 8407810 - List of Supplementary Quality Assurance Provisions, Tripod, Rifle Mount, 106MM: M27
LSQAP 8407830 - List of Supplementary Quality Assurance Provisions, Mount, 106MM Rifle: M92
DL 8403200 - Index of Inspection Lists, Mount, Rifle, 106MM: M79
DL 8407830 - Index of Inspection Lists, Mount, Rifle, 106MM: M92
DL 8773157 - Index of Inspection Lists, Tripod, Rifle Mount: M27
PDS P8407810 - Packaging Data Sheet, Tripod, Rifle Mount, 106MM Recoilless: M27
PDS P8407830 - Packaging Data Sheet, Mount, 106MM Rifle, Recoilless: M92
PDS P8447027 - Packaging Data Sheet, Rifle, Recoilless, 106MM, M40A2, W/E (Mount M79)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 First article. When specified in the contract, the contractor shall submit a first article. Unless otherwise specified (see 6.2.1), the first article shall be included in the first article package (see 5.1).

3.2 General requirements. Mounts, tripods and parts thereof shall conform to the drawings and specifications listed on Engineering Parts List EPL 8403200 for Mount M79, EPL 8407830 for Mount M92, EPL 8407810 for Tripod M27, and the requirements of this specification and MIL-C-13931.

3.3 Government-furnished equipment. Unless otherwise specified, one 106MM rifle and accessories will be furnished to a contractor who is supplying a complete mount for use in ascertaining the coordination of the rifle with the mount or tripod. Functional tests shall be made with the rifle in place and with a dummy round of ammunition or equivalently distributed weight in the chamber.



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3.4 Functioning. All parts shall function without binding or interference.

3.4.1 Base (Type I mount). The arm lock shall lock the arms in the spread and carrying positions, permitting no more than 1/4 inch horizontal play in either arm, measured at the end of the arm. The front arm wheel shall turn without binding or roughness when the mount is maneuvered on the ground. The base locking clamps shall lock the movable arms securely to the 4 x 4, 1/4 ton utility truck: M151A1C.

3.4.2 Tripod (Type II mount). The leg locks shall lock the legs in the open and folded positions, permitting no more than 3/16 inch horizontal play in either leg, measured at the end of the leg. The assembled tripod shall be capable of supporting a static load of 1000 pounds in the open position. The top of the tripod seat for the mount adapter and the mount adapter when seated shall be parallel with a level base within 1/16 inch, before and after a load test.

3.4.3 Rifle mounting. The 106MM rifle shall be capable of being mounted by placing the lug of the quick-breakdown sleeve of the rifle between the firing transfer-housing and the elevating cradle stud. Turning the elevating cradle locking yoke clockwise shall lock the rifle securely.

3.4.4 Traversing torque. The torque required to operate the traversing drive mechanism at constant speed shall be between 16 and 22 pound-inches. Traversing torque shall be adjusted by the use of shims between the cover and the clamp ring in the traversing drive assembly. Increasing shim thickness decreases torque and decreasing shim thickness increases torque.

3.4.4.1 Free traverse. When the free traverse knob is in the raised position, the traversing assembly shall be free to rotate without the use of the handwheel. Lowering the knob shall engage the drive mechanism.

3.4.4.2 Traversing stability. The force required to cause slippage in the traversing mechanism shall be 10.0 pounds or greater when measured under the following conditions.

a. Rifle at zero degrees elevation

b. Dummy round of ammunition in the rifle chamber

c. Traversing drive mechanism in the engaged position

d. Force applied at muzzle within one inch from the end of the tube and at right angles to and in the same horizontal plane as the rifle tube

e. Rifle at any position (angle) of traverse

NOTE: The force required to cause slippage varies directly with the traversing torque (see 3.4.4).



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3.4.5 Elevating mechanism. The elevating handwheel shall operate to elevate and depress the rifle between -5° depression and +30° elevation. The characteristic motion in depression is a discontinuous one, depending upon distribution of weight and speed of operation. Free motion of the muzzle in excess of 7/8 inch is an indication of excessive backlash. Torque shall not exceed 15 pound-inches on the vernier elevating knob or 60 pound-inches on the elevating handwheel.

3.4.6 Firing mechanism. The vernier elevating knob also operates the firing mechanism through the firing transfer tang. A pull of the knob fires the spotting rifle and a push fires the major weapon (106MM rifle). There shall be lines engraved on the firing transfer housing showing operating positions of the tang, both firing (major weapon and spotting rifle) and neutral. The tang shall show full movement between the lines representing firing positions when the firing knob is operated in both directions, and shall return to the neutral position after each operation. The tang shall engage to fire the rifle when the rifle is assembled to the mount.

3.5 Lubrication. All bearing and mating surfaces inside the elevating and traversing mechanisms shall be coated with a thin film of grease conforming to MIL-G-21164.

3.6 Performance.

3.6.1 Firing-resistance. Mounts and tripods shall be capable of withstanding the forces of firing when tested as specified in 4.5.7. There shall be no impairment in the functioning of the mount components during or after testing.

3.6.2 Material soundness. After testing, all parts shall be free from cracks and fractures and shall comply with soundness requirements specified on the applicable drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used in this specification are in accordance with MIL-STD-109.



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4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.4 First article inspection. A first article shall be submitted for inspection in accordance with contract requirements. The first article shall be representative of the production process to be used during quantity production. The first article shall be subjected to the inspections, examinations, and tests specified herein and such other inspections, examinations, and tests as are necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

4.5.1 Inspection provisions. Mounts shall be inspected for conformance to requirements of the provisions of this specification, MIL-C-13931, SQAPs listed in Master LSQAP 8403200, Master LSQAP 8407810, Master LSQAP 8407830, and SQAP-APPENDIX-WVT.

4.5.1.1 Final examination. Final examination of each mount shall be completed after testing and prior to preservation and packaging to determine compliance with this specification, and MIL-C-13931, and as specified in the contract (see 6.2.2).

4.5.1.2 Inspection approval stamp. The application of the inspection approval stamp shall be as specified in MIL-C-13931.

4.5.2 Inspection equipment. Data lists containing drawing numbers of Government-designed inspection equipment used in the inspection of parts, sub-assemblies and assemblies are listed in DL 8403200, DL 8407830, and DL 8773157.

4.5.2.1 Acquisition, maintenance, and disposition. Unless otherwise specified (see 6.2.1), responsibility for acquisition, maintenance, and disposition of acceptance inspection and test equipment prescribed on the List of Inspection Equipment, DL 8403200, DL 8407830, and DL 8773157, and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.2.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.



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4.5.3 Materials. Materials for use in component parts shall be inspected in accordance with MIL-C-13931 and the applicable material specifications referenced on the part drawing.

4.5.4 Parts. Parts, subassemblies, and assembled mounts and tripods shall be inspected in accordance with MIL-C-13931 and the applicable Supplementary Quality Assurance Provisions (SQAPs).

4.5.5 Examination.

4.5.5.1 Lubrication. The Government representative shall witness assembly of the elevating and traversing mechanisms to the extent necessary to assure compliance with the lubrication requirements of 3.5. The manufacturer shall furnish the Government representative with certification that the grease used in the assembly conforms to MIL-G-21164.

4.5.5.2 Complete assemblies. Mounts and tripods shall be assembled only from parts which have met the requirements, either after sampling inspection or screening. Defective pieces found during assembly shall be segregated for disposition. Each completed assembly shall be examined by the supplier for completeness and for verification of markings.

4.5.5.3 Workmanship. Workmanship shall be examined visually in accordance with MIL-C-13931.

4.5.6 Tests.

4.5.6.1 Functioning tests. Functioning tests shall be conducted in accordance with MIL-C-13931 and as specified herein. A minimum of two complete cycles of operation shall be performed on each mount.

4.5.6.2 Base (type I mount). The supplier shall operate each base without the rifle mounted, and maneuver the mount on the ground to determine conformance with 3.4.1. Failure of the arm lock to function, binding or excessive play in the arms, binding or roughness in the wheel or failure of the base locking clamps to function properly shall be cause for rejection.

4.5.6.3 Tripod (type II mount). The supplier shall operate each tripod to determine conformance with 3.4.2. The static load test shall be performed on each of the first ten tripods produced on a contract, and thereafter on two out of every 25 tripods. The test may be made in an arbor press equipped with indicating dial, or by another method which is approved by the Government representative. In each case, the static load test, when performed, shall be made just prior to the test of the requirement for parallelism. This shall be performed on each tripod by placing the legs, fully opened, on a surface previously determined to be level. The requirement for parallelism (3.4.2) shall then be checked with a spirit level, placed on the adapter seat in two positions at right angles, then on the adapter. If the level indicates a deviation of more than 1/16 inch in any position, the tripod or adapter shall be rejected.



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4.5.6.4 Rifle mounting. The tests of 4.5.6.5 and 4.5.6.6 shall be performed with the rifle mounted in the cradle. Conformance with the requirement of 3.4.3 shall be checked at the time of the mounting. Interference of any part, or failure of the yoke to lock the rifle securely shall be cause for rejection.

4.5.6.5 Traversing torque. Each mount shall be inspected with a rifle mounted and a complete round of dummy ammunition in the chamber, or with equivalently distributed weight attached, to assure conformance to the range prescribed in 3.4.4. Torque shall be measured with a spring scale attached to a strap wrapped around the traversing handwheel. Torque equals the tangential force required to turn the handwheel multiplied by the radius of the handwheel. An alternate method of measuring torque shall employ a torque wrench having an adapter to grasp the handwheel. Failure to conform with any requirements of 3.4.4 shall be cause for rejection.

4.5.6.6 Free traverse. Each mount with rifle in place as described in 4.5.6.5 shall be tested for conformance with the requirements of 3.4.4.1. The free traverse knob shall release the handwheel without binding. Failure to conform with any requirements of 3.4.4.1 shall be cause for rejection.

4.5.6.7 Traversing stability. The traversing stability shall be tested by using a suitable scale-type indicating device under the conditions specified in 3.4.4.2. Failure to conform with the limits of 3.4.4.2 shall be cause for rejection.

4.5.6.8 Elevating mechanism. With a complete round of dummy ammunition in the chamber of the rifle or with equivalently distributed weight attached, the elevating mechanism shall be operated from -5° to +30° (see 3.4.5). The torque on elevation shall be measured on both the vernier elevating knob and the elevating handwheel. Torque shall be measured with a spring scale attached to a strap wrapped around the vernier elevating knob or elevating handwheel as applicable. An alternate method of measuring torque shall employ a torque wrench having adapters to grasp the knob or handwheel as applicable. The free motion of the muzzle during depression shall be measured with a scale. Failure to conform with any requirements of 3.4.5 shall be cause for rejection.

4.5.6.9 Firing mechanism. The operation of the elevating knob to fire shall be tested by the supplier on each mount, before and after the assembly of the rifle. Failure to conform to any requirements of 3.4.6 shall be cause for rejection.

4.5.7 Performance testing.



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4.5.7.1 Proof firing. The first ten (10) mounts or tripods manufactured under a contract shall be assembled with both major and minor caliber rifles and function fired as complete systems. Firing for the major caliber rifle shall be as outlined in Table I. The standard rounds employed shall be as defined in 6.5. Proof firing tests of the M79 mount shall be repeated after completion of the road test and subsequent maneuvering test (see 4.5.7.3.2).

TABLE I

| No. of rounds* | Type of round (percent pressure) | Elevation |
|---|---|---|
| 1 | Standard (100) | 0° |
| 1 | Standard (100) | 30° |

*Additional standard rounds may be fired when, in the judgment of the proof director, the results of the previous rounds are considered questionable or inconclusive.

4.5.7.2 Road test. The first ten M79 mounts manufactured under a contract shall be assembled with rifle and mounted on a 4 by 4, 1/4 ton utility truck, M151A1C, and shall be given a road test. The road test shall consist of driving over unpaved rough roads or across country a minimum distance of 25 miles at varying speeds consistent with the capabilities of the truck.

4.5.7.3 Inspection after tests.

4.5.7.3.1 Visual examination. Following proof firing and road tests, the mounts and tripods shall be visually examined for security of assembly and for detection of deformation, cracks, or other defects that may have developed. The security of the mount (M79) to the vehicle shall be checked during and immediately following road test.

4.5.7.3.2 Maneuvering test. Each mount or tripod shall be maneuvered under simulated service to determine that all mechanisms and locking devices function properly.

4.5.7.3.3 Results of proving ground inspection. Results of tests and examinations shall be forwarded to the contracting officer for evaluation. Mounts or tripods rejected at the proving ground shall be adequately tagged so as to identify the deficiencies and returned to the supplier for correction of deficiencies.

4.5.8 Inspection of preparation for delivery.



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4.5.8.1 Packaging inspection. Unless otherwise specified, inspection to determine compliance with cleaning, preservation, packaging, packing, and marking requirements of the applicable packaging data sheets for the level designated in the contract shall be as specified in MIL-P-14232.

5. PREPARATION FOR DELIVERY

5.1 First article package. On each contract, a first article package shall be provided in accordance with 3.1. The first article package for the mount shall be unit packaged for the level of protection specified in the contract and packed level C. Unit packaging and packing shall be in accordance with MIL-P-14232 and Packaging Data Sheets P8707810, P8407830 and P8447027. Marking shall be in accordance with MIL-STD-129.

5.2 Unit packaging, packing and marking of cannon. The cannon shall be unit packaged and packed in accordance with the requirements of MIL-P-14232 and the applicable packaging data sheets (see 5.1), for the level of protection specified in the contract (see 6.2). Marking shall be in accordance with MIL-STD-129.

5.3 Unit packaging, packing and marking of repair parts. Repair parts shall be unit packaged and packed in accordance with the requirements of MIL-P-14232 and the applicable packaging data sheets (see 5.1), for the level of protection specified in the contract (see 6.2). Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Type I. The mount M79 is intended as a ground mount for the Cannon, 106MM Recoilless Rifle: M206. It can also be mounted on the 4 x 4, 1/4 ton utility truck: M151A1C.

6.1.2 Type II. The mount M92 is intended for mounting the Cannon, 106MM Recoilless Rifle: M206 on the Utility Platform Truck M274, with a tripod M27 furnished for ground mounting. The mount and tripod are also intended as a ground mount for the two upper outboard M206A1 cannons of the 106MM Multiple Recoilless Rifle M50.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements:

a. Title, number and date of the specification.



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Procurement requirements cont'd.

b. Requirements for submission of and shipping instructions for first article (see 3.1 and 4.4) and first article package. (See 3.1 and 5.1.)

(1) The contract should designate the examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

(2) The contract should specify the sample size (i.e., the number of units) of the first article.

(3) The contract should designate the inspection and test data that is required to be furnished the Government when the contractor is required to perform either or both examinations and tests.

(4) The contract should designate at whose expense a retest may be performed.

(5) The contract should specify that the acceptable first article shall be delivered in accordance with the terms of the contract, or

(6) Any additional or extended examinations and tests beyond the scope of this specification should be specified and identified as a separate item of the contract.

c. Block of serial numbers for the mounts.

d. List of inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.5.2).

e. Availability of inspection equipment from the Government.

f. Extent of supplier's responsibility for Government-furnished and for supplier-required final inspection equipment.

g. The levels of preservation, packaging, and packing required (see 5).

h. Packaging instructions for repair parts (see 5.3).

6.2.2 Contract data requirements. Monthly reports of the results of final examination and performance testing shall be as specified for delivery on DD Form 1423 in the contract.



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6.3 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control Systems Summary List, DD Form 1660.

6.4 Unless otherwise specified, the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

6.5 Definition. A standard pressure round is one employing a standard weight projectile and loaded to give a muzzle velocity of 1650 feet per second.

6.6 International standardization agreement. The mount M79 covered by this specification is the subject of international standardization agreement ABCA-ARMY-STD-46. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodian:
Army--WV

Preparing activity:
Army--WV

Project No. 1015-0260



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